Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is rated LOWER priority by elicitation, and the structural alignment is moderate. The benchmark is text-only Twitter content [Q38] matching the deployment's text-only modality, and Argentine professional cohorts have high digital infrastructure access [WEB-3, WEB-5]. Tokenization, URL/punctuation/user-id removal [Q38], and near-duplicate filtering at 0.75 cosine similarity [Q39, Q40] are reasonable preprocessing choices. However, three concerns remain: (a) the deployment spans Twitter/X, news articles, and press releases, but the benchmark is exclusively tweet-derived with no cross-channel transfer evaluation [WEB-20]; (b) language tagging is done via Google Cloud API [Q44, Q45] but lang_conf is missing for many CrisisMMD/AIDR examples (Concern 9), weakening language-stratified routing for the bilingual deployment; (c) all classification experiments are English-only [Q59] despite multilingual content being present in the consolidated data [Q42, Q43], so the form-level multilingual capability is not validated. The 70/10/20 split [Q60] and aggressive duplicate filtering [Q41, Q97] are methodological strengths.

ID	Evidence
Q38	“We first tokenize the text before applying any filtering. For tokenization, we used a modified version of the Tweet NLP tokenizer (O'Connor, Krieger, and Ahn 2010). Our modification includes lowercasing the text and removing URL, punctuation, and user id mentioned in the text. We then filter tweets having only one token.” (p.4)
Q39	“We then use a similarity-based approach to remove the near-duplicates. To do this, we first convert the tweets into vectors using bag-of-ngram approach as a vector representation. We use uni- and bi-grams with their frequency-based representations. We then use cosine similarity to compute a similarity score between two tweets and flag them as duplicate if their similarity score is greater than the threshold value of 0.75.” (p.4)
Q40	“To determine a plausible threshold value, we manually checked the tweets in different threshold bins (i.e., 0.70 to 1.0 with 0.05 interval) as shown in Figure 1, which we selected from consolidated informativeness dataset. By investigating the distribution and manual checking, we concluded that a threshold value of 0.75 is a reasonable choice.” (p.4)
Q41	“As indicated in Table 1, there is a drop after filtering, e.g., ~25% for informativeness and ~20% for humanitarian tasks. It is important to note that failing to eradicate duplicates from the consolidated dataset would potentially lead to” (p.4)
Q42	“Some of the existing datasets contain tweets in various languages (i.e., Spanish and Italian) without explicit assignment of a language tag.” (p.5)
Q43	“In addition, many tweets have codeswitched (i.e., multilingual) content.” (p.5)
Q44	“We decided to provide a language tag for each tweet if it is not available with the respective dataset.” (p.5)
Q45	“We used the language detection API of Google Cloud Services for this purpose.” (p.5)
Q59	“Although our consolidated dataset contains multilingual tweets, we only use tweets in English language in our experiments.” (p.6)
Q60	“We split data into train, dev, and test sets with a proportion of 70%, 10%, and 20%, respectively, also reported in Table 5.” (p.6)
Q97	“It is important to emphasize the fact that the results reported in this study are reliable as they are obtained on a dataset that has been cleaned from duplicate content, which might have led to misleading performance results otherwise.” (p.9)
WEB-3	88.4% Argentine national internet penetration as of January 2024 (https://datareportal.com/reports/digital-2024-argentina)
WEB-5	AMBA professional cohort fully connected; 71% of fixed-line subscriptions in CABA/Buenos Aires province/Córdoba/Santa Fe/Mendoza (https://freedomhouse.org/country/argentina/freedom-net/2024)
WEB-20	TREC CrisisFACTS extends to multi-stream (Twitter, Reddit, Facebook, online news) but provides no tweet-to-news transfer benchmarks (https://crisisfacts.github.io/)

Strengths

- Text-only Twitter modality matches the deployment's text-only requirement; no signal-distribution mismatch at the modality level
- Rigorous near-duplicate filtering (0.75 cosine threshold, manually validated) reduces overestimated test results [Q39, Q40, Q97]
- Language tags assigned to all tweets via Google Cloud API enable downstream language-stratified analysis [Q44, Q45]

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Q39	"We then use a similarity-based approach to remove the near-duplicates. To do this, we first convert the tweets into vectors using bag-of-ngram approach as a vector representation. We use uni- and bi-grams with their frequency-based representations. We then use cosine similarity to compute a similarity score between two tweets and flag them as duplicate if their similarity score is greater than the threshold value of 0.75." (p.4)
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Twitter text signal distributions match between benchmark and Argentine Twitter/X deployment at the form level. However, Twitter/X platform conventions have evolved post-2022 (post-Musk acquisition); pre-2017 tweet structure may differ in hashtag conventions, link formats, and length. INSUFFICIENT DOCUMENTATION on whether benchmark tokenization handles current X.com URL patterns.

IF-2: Argentine infrastructure supports text capture: 88.4% national internet penetration [WEB-3]; AMBA professional cohort has full connectivity [WEB-5]. No infrastructure obstacle.

IF-3: Cross-channel form gap: benchmark is tweet-only [Q38]; deployment also covers news articles and press releases. No cross-channel transfer benchmark exists in the literature [WEB-20]. Tweet-length constraints and Twitter-specific preprocessing (user-id removal, URL stripping) do not natively support longer-form institutional content.

IF-4: Documented form mismatches: (a) tweet-only vs. multi-channel deployment; (b) lang_conf missing for significant CrisisMMD/AIDR subset (Concern 9), reducing language-routing reliability; (c) temporal gap to current X.com platform conventions. These threaten external validity for non-Twitter deployment channels.

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Information Gaps

- Whether benchmark-trained classifiers transfer to news article and press release form factors with any usable F1
- Whether lang_conf NA examples disproportionately misclassify by language

Requires Expert Verification

- Operational assessment of whether Argentine deployment teams can downstream-restrict input to tweets only or must accept news/press content

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Remediation

Gap 1: Tweet-only benchmark does not cover deployment's news article and press release channels; lang_conf missing for CrisisMMD/AIDR subsets weakens language routing

Recommendation: Conduct cross-channel transfer evaluation by testing CrisisBench-trained classifiers on a held-out Argentine news article and press release corpus. Backfill lang_conf scores for affected examples using a current language-identification model. Consider TREC CrisisFACTS [WEB-20] as a multi-stream framework reference.

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